# Simple recursion on the ordinals: any step rules for successors and limits define a class function sigma on all ordinals

The Open Logic Project contributors

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the xi case display, the schema invocation and the three verification lines appear verbatim in the span.

*Editorial note.* The label simplerecursionschema names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own thm environment in the unit source.

*Allowed standard prerequisites (editorial).*

- Transfinite recursion schema. Same unit, \ollabel{transrecursionschema} (source line 744). Line 797: produces sigma with sigma(alpha) = xi of the restriction, for every ordinal alpha.
- Simple transfinite induction. Unit olp-set-theory-ordinals, \ollabel{simpletransrecursion} (source line 934 of that unit). Line 799: the verification of the three defining equations proceeds by zero/successor/limit induction.

*Source reference.* Original reference transrecursionschema: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 744–744. Prerequisite pointer invoked at proof line(s) 800; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference simpletransrecursion: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 948–948. Named cross-unit citation: the target label is staged in the corpus under unit olp-set-theory-ordinals at source line 948; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

**Theorem 0.1** (Simple Recursion)**.** *For any terms $\tau(x)$ and $\theta(x)$ and any set $A$, we can explicitly define a term $\sigma(x)$ such that:*

$$\begin{aligned} \sigma(\emptyset) &= A \\ \sigma(+1\alpha) &= \tau(\sigma(\alpha)) && \text{for any ordinal } \alpha \\ \sigma(\alpha) &= \theta(\operatorname{ran}\sigma \restriction \alpha) && \text{when } \alpha \text{ is a limit ordinal} \end{aligned}$$

*Proof.* We start by defining a term, $\xi(x)$, as follows:

$$\xi(x) = \begin{cases} A & \text{if } x \text{ is not a function whose} \\ & \quad \text{domain is an ordinal; otherwise:} \\ \tau(x(\alpha)) & \text{if } \operatorname{dom} x = +1\alpha \\ \theta(\operatorname{ran} x) & \text{if } \operatorname{dom} x \text{ is a limit ordinal} \end{cases}$$

By transrecursionschema, there is a term $\sigma(x)$ such that $\sigma(\alpha) = \xi(\sigma \restriction \alpha)$ for every ordinal $\alpha$; moreover, $\sigma \restriction \alpha$ is a function with domain $\alpha$. We show that $\sigma$ has the required properties, by simple transfinite induction (simpletransrecursion).

First, $\sigma(\emptyset) = \xi(\emptyset) = A$.

Next, $\sigma(+1\alpha) = \xi(\sigma \restriction +1\alpha) = \tau(\sigma \restriction +1\alpha(\alpha)) = \tau(\sigma(\alpha))$.

Last, $\sigma(\alpha) = \xi(\sigma \restriction \alpha) = \theta(\operatorname{ran}\sigma \restriction \alpha)$, when $\alpha$ is a limit. □